



# 清华大学

1. a)

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 |    | 1  | 1  | 1  |
| 1 |    |    | 1  | 1  |

$$F(x, y, z) = y$$

b)

|    | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 |    |    |    |    |
| 01 |    |    |    |    |
| 11 | 1  | 1  |    | 1  |
| 10 | 1  |    |    | 1  |

$$F(w, x, y, z) = \cancel{x'y + wxy'} \\ = x'y + wxy'$$

2. a)

|   | 00 | 01 | 11 | 10 |
|---|----|----|----|----|
| 0 | 1  | 1  | X  | X  |
| 1 | 1  | 1  | X  | 1  |

取  $X=1$ , 则  $F(x, y, z) = 1$

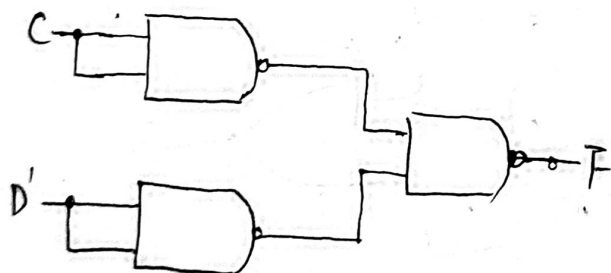
b)

|    | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 |    |    | X  |    |
| 01 |    | 1  | 1  | 1  |
| 11 | 1  | X  | 1  | 1  |
| 10 |    | X  | X  |    |

~~取  $X=1$ , 则  $F(x, y, z) =$~~

取  $X=1$ ,  $F(A, B, C, D) = BD + BC + AD + AB + CD$

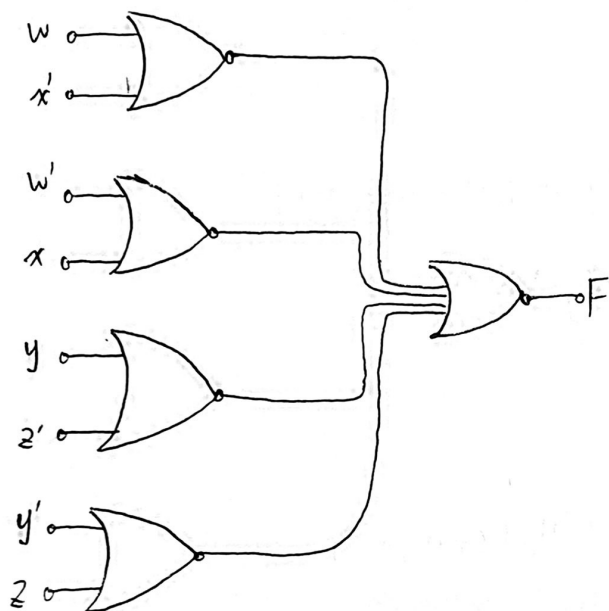
3.  $F(A, B, C, D) = \cancel{(A+A')}C'D' + (A' + AB + AB')C = C'D' + C = C + D'$



4.  $F(w, x, y, z) = \sum (0, 3, 12, 15) = \prod (1, 2, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14)$

|    | 00 | 01 | 10 | 11 |
|----|----|----|----|----|
| 00 |    | 0  | 0  | 0  |
| 01 | 0  | 0  | 0  | 0  |
| 10 |    | 0  |    | 0  |
| 11 | 0  | 0  | 0  | 0  |

~~F(x,y)~~,  $F(w, x, y, z) = (w+x)(w+x)(y+z')(y+z)$

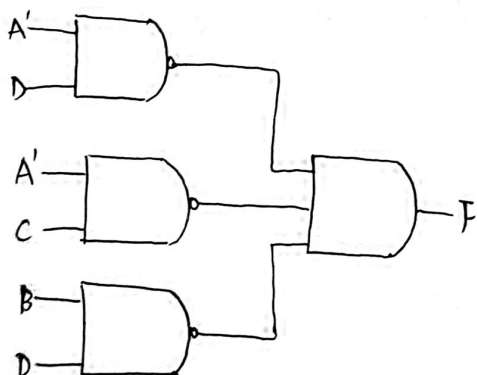


5.  $F(A, B, C, D) = \sum (0, 4, 8, 9, 10, 11, 12, 14) = \prod (1, 2, 3, 5, 6, 7, 13, 15)$

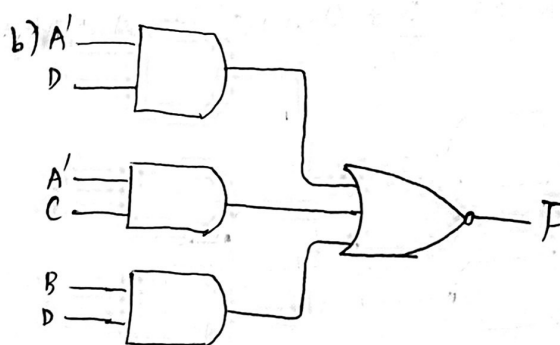
|    | 00 | 01 | 11 | 10 |
|----|----|----|----|----|
| 00 | 1  | 0  | 0  | 0  |
| 01 | 1  | 0  | 0  | 0  |
| 11 | 1  | 0  | 0  | 1  |
| 10 | 1  | 1  | 1  | 1  |

$F(A, B, C, D) = \cancel{C'D'} + AB' + AD' = \cancel{AD' + AC'} + \cancel{AC}$   
 $= \cancel{AC}$   
 $= (A+D')(A+C')(B'+D')$

a)

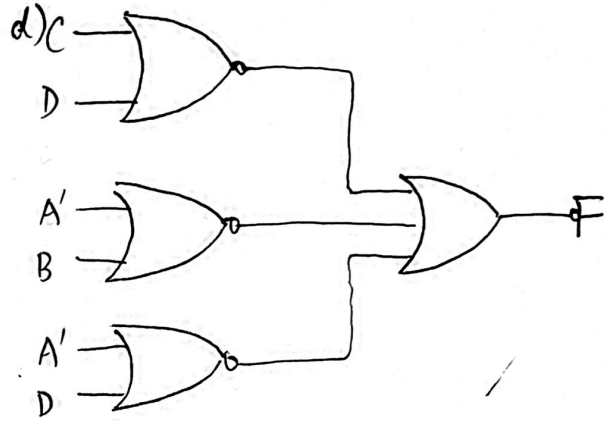
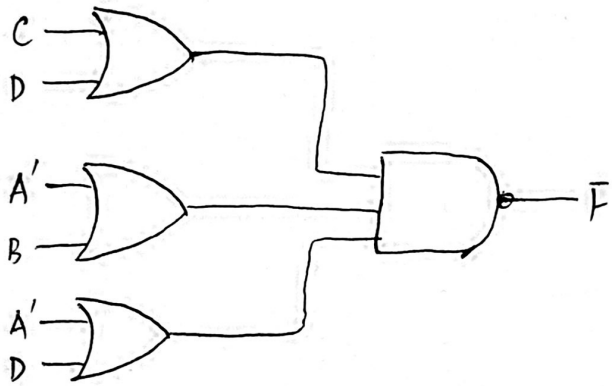


b)





c)



b.  $F(a, b, c, d, e, f) = \overline{abc + de + f} = \overline{a + b + c + d + e + f}$   
 $= \overline{abc} \cdot \overline{de} \cdot \overline{f}$

